Md. Jubaer Khan

Senior Software Engineer — Embedded Systems & C/C++

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PROFESSIONAL SUMMARY

Senior Software Engineer with over five years of experience delivering embedded, desktop and web software, including four-plus years at BJIT Ltd. building safety-critical systems for Japanese and international clients. Strong in C and C++ across RTOS and Embedded Linux targets, automotive and industrial communication protocols (CAN 2.0B, SAE J1939), and MISRA-C / MISRA-C++ / BARR-C compliant development. Experienced in requirement analysis, system architecture, code review and mentoring within Agile/Scrum teams. Recent work applies AI agents across the software development life cycle and modernises legacy VB6 systems to web-based C#/.NET. Recognised as BJIT Best Performer of the Year for 2023–2024, and co-author of a peer-reviewed IEEE Transactions paper on deep-learning-based COVID-19 detection.

TECHNICAL SKILLS

Languages	C, C++ (Embedded C/C++, Visual C++), Python, C#, VB.NET, Java (core), JavaScript
Embedded & RTOS	RTOS, Embedded Linux (Yocto, Buildroot), STM32 and ESP32/ESP8266 ecosystems, embedded system design, peripheral and sensor interfacing, schematic and datasheet analysis
Protocols	CAN 2.0B, SAE J1939, UART, SPI, I2C, RFID, Wi-Fi (IEEE 802.11)
Standards	MISRA-C, MISRA-C++, BARR-C, design patterns, UML, OOAD, test-driven development, code and design review
Desktop & Web	Qt Framework, MFC, .NET Framework / .NET Core, ASP.NET MVC, Razor, Flask, Django, HTML, CSS, Bootstrap, MySQL, MS SQL Server
AI & ML	AI agents and agentic workflows, machine learning and NLP fundamentals (TensorFlow, Keras, CNN)
Tools & Process	Git, GitHub, Bitbucket, Jira, Confluence, Visual Studio, VS Code, PlatformIO, CodeWarrior, Orchestra Suite, Agile/Scrum

PROFESSIONAL EXPERIENCE

Senior Software Engineer *Jul 2025 – Present*

BJIT Ltd. • Dhaka, Bangladesh

- Provide technical leadership for embedded and PC application projects, owning system architecture and critical modules.
- Lead design and code reviews, enforcing MISRA-C, MISRA-C++ and BARR-C compliance across project teams.
- Mentor junior and mid-level engineers through design walkthroughs, technical guidance and best-practice sharing.
- Resolve complex system-level defects in RTOS and Embedded Linux environments and support integration phases.

Software Engineer *Apr 2022 – Jun 2025*

BJIT Ltd. • Dhaka, Bangladesh

- Designed, implemented, tested and deployed embedded, desktop and web software components for client products.
- Developed communication-driven applications over CAN 2.0B, SAE J1939, UART, SPI and I2C.
- Built custom Embedded Linux distributions with Yocto and Buildroot for target hardware platforms.
- Delivered cross-platform desktop applications in C++/Qt and PC applications on .NET (C#, VB.NET).

Web Developer & Designer (part-time) *Nov 2021 – Feb 2022*

Independent client engagement

- Planned, built, deployed and maintained a client business website with WordPress and cPanel, including professional email setup.

Intern Software Engineer *Feb 2021 – Jun 2021*

North South University — Office of Information Technology

- Led a three-member team to build an NLP-based student assistance chatbot in Python and Flask, deployed live on university infrastructure.
- Led development of an RFID-based automated printing and access facility using C++, ESP32 and the RC522 reader over SPI.

Student Worker, IT Support Team *Aug 2018 – Jan 2021*

North South University — Office of Information Technology

- Provided technical support to faculty, officials and students, and staffed the IT service desk and request tracking.
- Prepared and maintained PCs for printing and engineering labs, covering OS and software installation, IP and domain setup.

SELECTED PROFESSIONAL PROJECTS

Client and product names are withheld under confidentiality agreements.

In-House R&D — Agentic SDLC & AI Agent-Based Monitoring *Jul 2026 – Aug 2026*
Two in-house research and development initiatives: applying AI agents across the software development life cycle (requirements, design, coding, testing and review), and AI-agent-based monitoring software. **Tech:** *AI agents, SDLC • In-house R&D*

Legacy Management Software Modernisation — VB6 to C#/.NET *Nov 2025 – Present*
Migration of a legacy VB6 management application to a web-based C#/.NET system, including analysis of existing functionality and re-implementation. **Tech:** *VB6, C#, .NET, ASP.NET, web*

Automotive Embedded Control Platform — 2nd generation *Feb 2025 – Nov 2025*
Design and development of safety-critical embedded modules for automotive-grade vehicle systems, from analysis and detailed design through production and test. Contributed to architecture discussions, code review and system integration. **Tech:** *C, C++, CAN 2.0B, SAE J1939, RTOS • Team of 5*

Internal R&D, PC Application OJT & Engineering Support Platforms *Nov 2023 – Present*
Development of internal tools, proofs of concept and training platforms supporting on-the-job training and experimental work across embedded and desktop stacks. **Tech:** *C#, C/C++, RTOS, Embedded Linux, Unity • Team of 19*

Custom Embedded Linux Distribution *Nov 2023 – Mar 2024*
Built and customised Embedded Linux images tailored to specific target hardware, including layer configuration, package selection and image tuning. **Tech:** *Yocto, Buildroot, Linux • Team of 2*

Cross-Platform Desktop Application (Qt) *Sep 2023 – Oct 2023*
Developed a high-performance cross-platform GUI application, covering requirement analysis, detailed design and implementation. **Tech:** *C++, Qt Framework • Team of 2*

Automotive Embedded Control Platform — 1st generation *Jun 2022 – Feb 2023*
Developed vehicle interface management software for an automotive client, covering analysis, detailed design, production and test under embedded coding standards. **Tech:** *C, C++, CAN 2.0B, Python • Team of 6*

EDUCATION

B.Sc. in Computer Science & Engineering *May 2017 – May 2021*
North South University, Dhaka, Bangladesh
Cum laude • Specialisation coursework: Machine Learning, Pattern Recognition & Neural Networks, Theory of Computation, Mobile & Wireless Application Development

Higher Secondary Certificate (HSC), Science *2013 – 2015*
Jhalokathi Govt. College, Jhalokathi, Bangladesh

RESEARCH & PUBLICATIONS

XCR-Net: A Computer Aided Framework to Detect COVID-19 *2024*
Ashik Mostafa Alvi, **Md. Jubaer Khan**, Nishat Tasnim Manami, Zubair Azim Miazi, Kate Wang, Siuly Siuly, Hua Wang • *IEEE Transactions on Consumer Electronics, vol. 70, no. 4, pp. 7551–7561* • doi.org/10.1109/TCE.2024.3446793

Building COVID Knowledge Base from Medical Journals *2021*
Md. Jubaer Khan, Walidul Alam Ehab • *Undergraduate thesis, Dept. of Electrical & Computer Engineering, North South University* • repository.northsouth.edu

AWARDS & ACHIEVEMENTS

Best Performer of the Year — BJIT Ltd. *2023–2024*

LANGUAGES & ADDITIONAL INFORMATION

Training	Embedded Systems engineering course, BJIT Academy (2022) • AI Fundamentals, IBM SkillsBuild (2026) • Information Security IS-101, BJIT Academy (2024)
Languages	Bangla — native • English — fluent (reading, writing, speaking)
Personal	Date of birth 13 December 1998 • Nationality: Bangladeshi

References available on request.